

Palpable Night Forest

A Sensorial Interface for Remote Engagement with Nocturnal Nature

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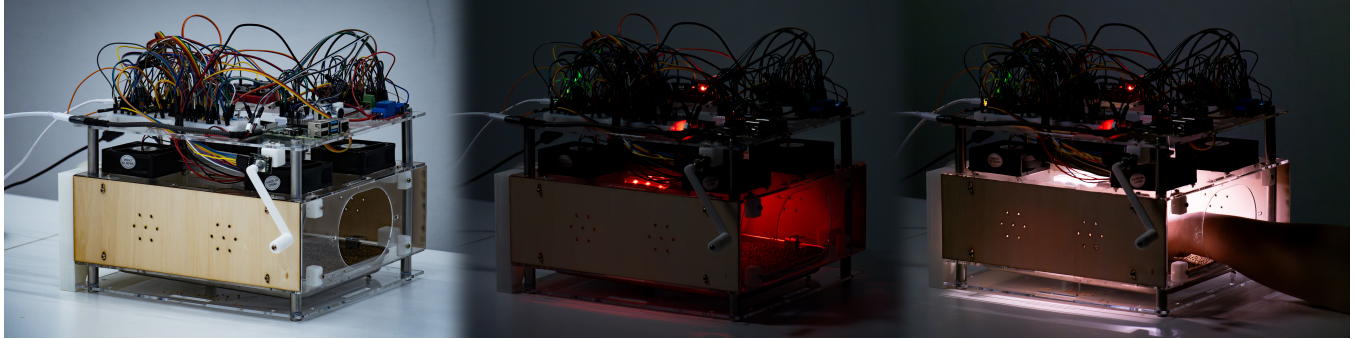


Figure 1: Left: Our system only operates at nighttime. Center: The machine is warming up its internal components. Right: The user inserts their hand deep and experiences a specific location's nighttime nature through atmospheric conditions.

Abstract

Humans and nature have become increasingly disconnected due to rapid urbanization. To reconnect humans with nature, HCI studies have explored ways of remotely engaging with distant natural environments. However, current research on remote human–nature engagement largely focuses on daytime or afternoon experiences involving vision and sound, often overlooking the unique qualities of nocturnal nature. In this research, we propose an interface that allows users to experience the distinct characteristics of nocturnal nature from distant indoor spaces. The contribution of this poster is twofold: 1) we introduce a specifically designed system that enables remote engagement with nocturnal natural environments; 2) we conducted a study using this system to identify potential improvements for a more effective human–nocturnal nature communication interface.

CCS Concepts

• **Hardware** → *Emerging interfaces*.

Keywords

Night, Nature, Sensory interfaces, Remote connections

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1 Introduction

Human and nature have been disconnected in the modern era—both physical separation and psychological alienation [13, 16]. Rapid urbanization has pushed nature far away from our daily lives. The fields and mountains that were once nearby are now difficult to reach for many city dwellers, disrupting the first-hand, bodily experiences that ground human–nature relationships [9, 12].

To reconnect humans with nature, recent Human–Computer Interaction (HCI) studies have sought to engage with remote natural environments [4, 8, 11], bringing wilderness into daily life to provide rich sensory experiences often lacking within urban constraints. These attempts include not only providing remote screens through user-deployed or user-built cameras [6] but also replicating natural phenomena like ocean waves as kinetic art [3]. Furthermore, some researchers have explored sensory embodiment, such as translating wind into a tactile experience on a garment [10].

While remote connections to nature have expanded to all times of day [2], systems capable of representing nocturnal environments still mainly rely on audiovisual means [4], such as live-streamed night-vision cameras [1]. However, these modalities are insufficient to convey the unique and rich experience of the dark wilderness—imagine walking through a dark mountain trail that feels eerie, or a humid, rainy night stroll with wind blowing across a sea-facing beach.

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In this context, we propose a different approach that enables users to remotely experience distant natural environments from indoor spaces by involving various modalities—tactile, auditory, olfactory—along with ritual qualities. We focus on non-visual approaches, as scholars such as Edensor [5] argue that restricted vision heightens other senses—like the feeling of air on the skin—creating a unique sensory richness, which we believe nocturnal nature inherently possesses. Additionally, as Griffiths and de Bézenac [7] note, efforts to connect humans with the non-visual, multisensory aspects of nocturnal nature remain limited, highlighting a compelling opportunity for exploration.

2 Understanding User Expectations

To understand what nocturnal qualities people expect and want to experience or engage with from a distance, we conducted an online survey with open-ended questions targeting university students with no prerequisite interest in nature ($n = 48$). We analyzed the responses using thematic analysis, identifying 99 codes that yielded three main themes—listed below—representing common user expectations for connecting with nocturnal nature.

1. Participants appreciated the unique sensory qualities of the night—such as coldness and dampness—as a distinct departure from the daytime experience.
2. They also perceived the night as a mysterious world that invites curiosity, expressing a desire to actively use their imagination and patience to explore the unseen.
3. Participants expressed a willingness to confront fear, unpredictability, or the unknown when considering experiences with dark, distant nature.

3 Implementation: ‘Palpable Night Forest’

To develop the nocturnal experiential system, we incorporated findings from the earlier preliminary study. Below, we share the user interaction flow and the architecture behind it.

3.1 Interaction Flow

To operate the machine, an individual user must first go through the following ritual-based preparatory procedures after powering it on:

A) The user must turn off the indoor room lights during nighttime in their geographical location. **B)** Once the lights are off, the device’s inner LED strips begin flashing red—blinking every 0.1 seconds in 3-second intervals. This signals that the user should now position themselves silently in front of the device in the dark room. **C)** Once the user is seated in front of the machine in silence and darkness, the red LEDs begin to glow steadily brighter (without blinking) for 15 seconds. **D)** This 15-second illumination indicates that the ritual has been successfully recognized by the machine, offering the user a sense of anticipation. It also signifies that the machine is warming up its internal components (see 3.2). **E)** After the 15-second preparation period, the glowing red LED strips turn white, signaling that the machine is ready to deliver the nocturnal experience once the user inserts their hand into the device.

We employed the ritual as a method, as it is known to enhance experiential awareness [14, 15]. We consider such enhancement necessary, as the expected nocturnal qualities identified in our study

involve not only various sensory modalities, but also feelings of curiosity and unpredictability.

Guided by the white LED (see E), the user inserts their hand deep into the device’s opening—a hole-shaped interface designed to allow a hand to enter. Once the hand is inserted, the user can experience a specific location’s nighttime nature—for example, a forest area near a rural village in a specific country—through atmospheric conditions, as the machine reads real-time environmental data and translates it into sensations such as coolness, humidity, airflow, touchable soil, and ambient sound that mimic the specific nocturnal environment. A side-mounted lever allows the user to shift or move between different locations, with the atmosphere changing in real time.

3.2 System Configuration

Our system ($310 \times 270 \times 270$ mm) is controlled by two microcontrollers: a Raspberry Pi-based server and an ESP32-based client. The server collects real-time atmospheric temperature, humidity, wind speed, and wind direction data from specific latitude and longitude coordinates via the *Open-Meteo* API. Simultaneously, it extracts audio streams from *YouTube* live cams to provide the location’s ambient sound. The client device recognizes ambient light (via CdS sensor), sound (via microphone), and proximity (via ultrasonic sensors), and receives user input for selecting a new remote location through a rotary encoder—the lever. It requests atmospheric data from the server via Wi-Fi and uses this information to control the actuators in real time: a quadruple fan array, a Peltier element for cooling below zero, and a misting device. These actuators determine how cool and humid the air should be, and where the breeze should flow, with LEDs indicating their operational status.

4 User Study for Potential Improvement

We identified potential areas for improving the device’s usability and enhancing the overall experiential quality of nighttime nature through a user study with six participants: three nighttime nature lovers—who enjoy light-trapping for insects, jogging alongside a dark stream, stargazing—and three interaction designers with bachelor’s degrees in design majors.

Participants pointed out the coolness and dampness were not pervasive enough. They wanted to feel its dampness also from the wet soil, not just through the humid air. Some even showed willingness to accept exaggerations—such as getting soaked by a heavy downpour.

When they moved between remote sites, participants found the shifts in sensory qualities to be subtle and nuanced, yet sometimes confusing at the same time. They suggested that providing geographical information would help contextualize the experience.

They also reported on detailed improvement in device usability: for example, the hand-insertion area was not spacious enough, and some found the device’s inner LED light too harsh or sharp.

5 Conclusion

In this poster, we introduced a method or interface for experiencing nocturnal nature from a distance—in this case, a nighttime forest. To develop the device, we conducted a preliminary survey and identified people’s expectations for the nocturnal qualities they

want to experience or engage with from afar. Through a user study using the developed machine, we identified key improvements for our remote human–nocturnal nature experience interface for further study. We hope our unusual interface provokes new ideas and discussions for those interested in researching ways of engaging with nature.

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